The project was a great success for Maureen. She was able to reduce the time needed to set up an enterprise application from months to hours because AWS provides virtual machines, databases, and even the networking infrastructure on demand within a few minutes. Maureen's project also benefits from lower infrastructure costs on AWS, compared to using its own infrastructure on-premises.

### *1.2.3 Implementing a highly available system*

Alexa is a software engineer working for a fast-growing startup. She knows that Murphy's Law applies to IT infrastructure: anything that can go wrong *will* go wrong. Alexa is working hard to build a highly available system to prevent outages from ruining the business. All services on AWS are either highly available or can be used in a highly available way. So, Alexa builds a system like the one shown in figure 1.5 with a high availability architecture. The database service is offered with replication and fail-over handling. In case the primary database instance fails, the standby database is promoted as the new primary database automatically. Alexa uses virtual machines acting as web servers. These virtual machines aren't highly available by default, but Alexa launches multiple virtual machines in different data centers to achieve high availability. A load balancer checks the health of the web servers and forwards requests to healthy machines.